Statement 1: Plot activation functions used in neural networks

This viva will mostly test your understanding of **what activation functions are, why they are needed, their types, formulas, and graph behavior.**

---

1. What is an activation function?

Answer:
An activation function is a mathematical function applied to the output of a neuron. It decides whether the neuron should be activated and introduces non-linearity into the neural network.

---

2. Why do we use activation functions?

Answer:
Activation functions help neural networks learn complex and non-linear patterns. Without them, the network behaves like a simple linear model and cannot solve complicated problems.

---

3. What happens if no activation function is used?

Answer:
Without activation functions, all layers behave like a single linear transformation. The neural network loses its ability to model non-linear relationships.

---

4. What is non-linearity in neural networks?

Answer:
Non-linearity means the relationship between input and output is not a straight-line function. It allows the network to solve real-world complex problems like image recognition.

---

5. Name common activation functions.

Answer:
Common activation functions are:

- Binary Step
- Sigmoid
- Tanh

- ReLU
- Leaky ReLU
- Softmax

---

6. What is Binary Step activation function?

Answer:
Binary Step gives output as either 0 or 1 depending on a threshold. It is mainly used in simple perceptrons.

Formula:

$$f(x) = \begin{cases} 1, x \geq 0 \\ 0, x < 0 \end{cases}$$

---

7. What is Sigmoid activation function?

Answer:
Sigmoid converts input into values between 0 and 1. It is commonly used in binary classification problems.

Formula:

$$f(x) = \frac{1}{1 + e^{-x}}$$

---

8. Advantages of Sigmoid?

Answer:
Sigmoid is smooth, differentiable, and maps outputs into probability values between 0 and 1.

---

9. Disadvantages of Sigmoid?

Answer:
It suffers from the vanishing gradient problem for large input values, making training slower in deep networks.

---

10. What is Tanh activation function?

Answer:
Tanh maps values between -1 and +1. It is zero-centered and often performs better than sigmoid.

Formula:

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

---

11. Difference between Sigmoid and Tanh?

Answer:
Sigmoid outputs between 0 and 1, while Tanh outputs between -1 and 1. Tanh is zero-centered, so it usually converges faster.

---

12. What is ReLU?

Answer:
ReLU (Rectified Linear Unit) outputs the input directly if positive, otherwise outputs zero. It is widely used in deep learning.

Formula:

$$f(x) = max(0, x)$$

---

13. Why is ReLU popular?

Answer:
ReLU is computationally simple, trains faster, and reduces the vanishing gradient problem.

---

14. What is the dead neuron problem?

Answer:
In ReLU, if a neuron always receives negative input, it outputs zero permanently and stops learning. This is called the dead neuron problem.

---

15. What is Leaky ReLU?

Answer:
Leaky ReLU allows a small negative output for negative inputs, solving the dead neuron problem.

Formula:

$$f(x) = \begin{cases} x, x > 0 \\ 0.01x, x \leq 0 \end{cases}$$

---

16. What is Softmax?

Answer:
Softmax converts outputs into probabilities whose sum equals 1. It is used in multi-class classification.

---

17. Which activation is used in hidden layers?

Answer:
ReLU is mostly used in hidden layers because it is fast and efficient.

---

18. Which activation is used in output layer?

Answer:

- **Sigmoid** → Binary classification
- **Softmax** → Multi-class classification
- **Linear** → Regression

Procedure / Algorithm for Statement 1

If examiner asks: **Explain procedure of plotting activation functions**

Answer:

Step 1:

Import required libraries like NumPy and Matplotlib.

Step 2:

Create input values over a range (for example -10 to 10).

Step 3:

Define mathematical functions for:

- Sigmoid
- Tanh
- ReLU
- Leaky ReLU

Step 4:

Compute output for each activation function.

Step 5:

Plot graphs using matplotlib.

Step 6:

Compare graph shapes and analyze their behavior.

Possible Cross Question

Why do we plot activation functions?

Answer:
Plotting helps visualize how each function transforms input values and understand their behavior during training.

Very Important Practical Viva Question

Which activation function is best?

Answer:
There is no universally best activation function. ReLU is commonly preferred for hidden layers, while output layer activation depends on the problem type.

One-liner for quick revision

Activation function introduces non-linearity in neural networks, enabling them to learn complex patterns.

Send **Statement 2** now (McCulloch-Pitts / ANDNOT).

yea

Statement 2: Generate ANDNOT function using McCulloch-Pitts Neural Network

This viva mainly focuses on:

- McCulloch-Pitts neuron
- Logic gates
- Thresholding
- Weights and bias
- Binary outputs

---

1. What is McCulloch-Pitts Neural Network?

Answer:
McCulloch-Pitts neuron is the simplest artificial neuron model proposed in 1943. It works on binary inputs and binary outputs using threshold logic.

---

2. Why is McCulloch-Pitts model important?

Answer:
It is the foundation of artificial neural networks and demonstrates how logical operations can be performed using neurons.

---

3. What are the characteristics of McCulloch-Pitts neuron?

Answer:

- Uses binary inputs (0 or 1)
- Produces binary output

- Uses weighted sum
- Uses threshold activation function

---

4. What is threshold in McCulloch-Pitts neuron?

Answer:
Threshold is the minimum value required for the neuron to activate and produce output 1.

---

5. What is the ANDNOT function?

Answer:
ANDNOT gives output 1 only when first input is 1 and second input is 0.

---

6. Truth table for ANDNOT gate?

X1	X2	Output
0	0	0
0	1	0
1	0	1
1	1	0

---

7. How is ANDNOT implemented using McCulloch-Pitts neuron?

Answer:
Positive weight is given to first input and negative weight to second input. The neuron activates only when X1 = 1 and X2 = 0.

---

8. What weights are generally used for ANDNOT?

Answer:
Usually:

- $w_1 = 1$
- $w_2 = -1$

Threshold = 1

---

9. Why is second weight negative?

Answer:
The second input inhibits neuron activation. If X2 becomes 1, it reduces the weighted sum and prevents output activation.

---

10. What activation function is used?

Answer:
Binary step activation function is used.

$$f(x) = \begin{cases} 1, x \geq \theta \\ 0, x < \theta \end{cases}$$

---

11. Explain working of ANDNOT for input (1,0).

Answer:
Weighted sum:

$$(1 \times 1) + (0 \times -1) = 1$$

Since sum equals threshold, output becomes 1.

---

12. Explain working for input (1,1).

Answer:
Weighted sum:

$$(1 \times 1) + (1 \times -1) = 0$$

Since sum is below threshold, output becomes 0.

---

13. What type of problems can McCulloch-Pitts neuron solve?

Answer:
It can solve linearly separable problems such as AND, OR, and ANDNOT gates.

---

14. What are limitations of McCulloch-Pitts neuron?

Answer:

- Cannot learn automatically
- Fixed weights
- Cannot solve non-linear problems like XOR

---

15. Difference between McCulloch-Pitts and Perceptron?

Answer:
McCulloch-Pitts uses fixed weights and no learning, while perceptron can learn weights using a learning algorithm.

Procedure / Algorithm

Algorithm for ANDNOT using McCulloch-Pitts

Step 1:

Take binary inputs X1 and X2.

Step 2:

Assign weights:

$$w_1 = 1, w_2 = -1$$

Step 3:

Calculate weighted sum:

$$Y_{in} = x_1 w_1 + x_2 w_2$$

Step 4:

Compare weighted sum with threshold.

If:

$$Y_{in} \geq 1$$

output = 1

Else:
output = 0

Step 5:

Repeat for all input combinations.

Important Viva Cross Question

Why is ANDNOT called linearly separable?

Answer:
Because its outputs can be separated using a straight-line decision boundary.

Another Important Question

Can McCulloch-Pitts neuron learn from data?

Answer:
No. It does not have a training mechanism or weight updating process.

Quick Revision Line

McCulloch-Pitts neuron is a binary threshold neuron used to implement basic logic gates using fixed weights and thresholds.

3

Statement 3: Perceptron Neural Network to Recognize Even and Odd Numbers

This viva mainly covers:

- Perceptron
- Classification
- Even/Odd recognition
- ASCII values

- Supervised learning

---

1. What is a Perceptron?

Answer:
Perceptron is the simplest supervised learning neural network used for binary classification problems. It consists of inputs, weights, bias, and an activation function.

---

2. Who developed the Perceptron?

Answer:
Perceptron was developed by Frank Rosenblatt in 1958.

---

3. What is the main purpose of a Perceptron?

Answer:
The main purpose of a perceptron is to classify input data into two categories using a decision boundary.

---

4. Why is this problem called binary classification?

Answer:
Because numbers are classified into only two classes:

- Even
- Odd

---

5. What is ASCII?

Answer:
ASCII (American Standard Code for Information Interchange) is a character encoding system where each character is represented by a numeric value.

---

6. Why are ASCII values used in this program?

Answer:
ASCII values provide numerical input representation for digits 0 to 9, which can be processed by the perceptron.

---

7. How are even and odd numbers identified?

Answer:
If a number is divisible by 2, it is even; otherwise, it is odd.

---

8. What are the components of a perceptron?

Answer:
Main components are:

- Input layer
- Weights
- Bias
- Summation function
- Activation function

---

9. What activation function is generally used in perceptron?

Answer:
Binary step activation function is commonly used.

$$f(x) = \begin{cases} 1, x \geq 0 \\ 0, x < 0 \end{cases}$$

---

10. What is the role of weights?

Answer:
Weights determine the importance of each input feature in prediction.

---

11. What is bias in perceptron?

Answer:
Bias helps shift the decision boundary and improves classification flexibility.

---

12. What is supervised learning?

Answer:
Supervised learning is a learning method where the model is trained using labeled input-output pairs.

---

13. What are target outputs in this problem?

Answer:
Usually:

- Even → 0
- Odd → 1

(or vice versa depending on implementation)

---

14. What is the perceptron learning rule?

Answer:
The perceptron learning rule updates weights based on prediction error.

$$w_{new} = w_{old} + \eta(t - y)x$$

where:

- η= learning rate
- t= target output
- y= predicted output

---

15. What is learning rate?

Answer:
Learning rate controls how much weights are updated during training.

---

16. What is an epoch?

Answer:
One complete pass through the entire training dataset is called an epoch.

---

17. What is decision boundary?

Answer:
Decision boundary is a line or hyperplane that separates different classes.

---

18. Can perceptron solve XOR problem?

Answer:
No. Perceptron can solve only linearly separable problems, while XOR is non-linearly separable.

---

19. Why can perceptron classify even and odd numbers?

Answer:
Because the training data can be separated into two classes using learned weights and thresholds.

---

20. What is linearly separable data?

Answer:
Data that can be separated using a straight line or hyperplane is called linearly separable data.

Procedure / Algorithm

Algorithm for Even/Odd Recognition using Perceptron

Step 1:

Take numbers from 0 to 9.

Step 2:

Convert numbers into ASCII values.

Example:

- '0' → 48
- '1' → 49

Step 3:

Assign target outputs:

- Even → 0
- Odd → 1

Step 4:

Initialize:

- Random weights
- Bias
- Learning rate

Step 5:

Compute weighted sum:

$$Y_{in} = wx + b$$

Step 6:

Apply activation function.

If:

$$Y_{in} \geq 0$$

output = 1

Else output = 0.

Step 7:

Compare predicted output with target output.

Step 8:

Update weights using perceptron learning rule.

Step 9:

Repeat until error becomes minimum.

Important Cross Questions

Why is perceptron called a single-layer neural network?

Answer:
Because it contains only one trainable layer of neurons and no hidden layers.

Difference between perceptron and ANN?

Answer:
Perceptron is a simple single-layer model, while ANN contains multiple layers and can solve complex non-linear problems.

Why are hidden layers important?

Answer:
Hidden layers help neural networks learn complex and non-linear patterns.

Very Important Viva Question

What are limitations of perceptron?

Answer:

- Solves only linearly separable problems
- Cannot solve XOR problem
- Limited learning capability

Quick Revision Line

Perceptron is a supervised binary classifier that learns weights using the perceptron learning rule to separate data into two classes.

4

Statement 4: Demonstrate Perceptron Learning Law with Decision Regions

This viva mainly focuses on:

- Perceptron learning algorithm
- Weight updating
- Classification
- Decision boundary / decision region
- Linear separability

---

1. What is Perceptron Learning Law?

Answer:
Perceptron Learning Law is an algorithm used to train a perceptron by updating weights based on prediction error.

---

2. Why is Perceptron Learning Law used?

Answer:
It helps the perceptron learn correct weights so that inputs can be classified accurately.

---

3. What is the formula for Perceptron Learning Law?

$$w_{new} = w_{old} + \eta(t - y)x$$

Where:

- η→ learning rate
- t→ target output
- y→ predicted output
- x→ input

---

4. What happens if prediction is correct?

Answer:
If predicted output equals target output, weights are not updated.

---

5. What happens if prediction is wrong?

Answer:
Weights and bias are updated to reduce classification error.

---

6. What is learning rate?

Answer:
Learning rate controls the amount of weight change during training.

---

7. What is a decision boundary?

Answer:
Decision boundary is a line or hyperplane that separates different classes in the input space.

---

8. What is a decision region?

Answer:
Decision region is the area in which all points belong to the same class after classification.

---

9. Why are decision regions important?

Answer:
They visually show how the perceptron separates different classes.

---

10. What is linear separability?

Answer:
If two classes can be separated using a straight line, they are linearly separable.

---

11. Can perceptron classify non-linearly separable data?

Answer:
No. Single-layer perceptron cannot solve non-linearly separable problems like XOR.

---

12. What is the role of weights in decision boundary?

Answer:
Weights determine the slope and position of the decision boundary.

---

13. What is bias in perceptron?

Answer:
Bias shifts the decision boundary to improve classification.

---

14. What activation function is used in perceptron?

Answer:
Binary step activation function is generally used.

---

15. What is convergence in perceptron learning?

Answer:
Convergence means the perceptron has learned weights that correctly classify all training samples.

---

16. When does perceptron learning stop?

Answer:
Training stops when:

- all samples are classified correctly, or
- maximum epochs are reached.

---

17. What is an epoch?

Answer:
One complete iteration through the entire training dataset is called an epoch.

---

18. Why is graphical representation used?

Answer:
Graphical representation helps visualize training samples and the decision boundary.

---

19. What does the perceptron actually learn?

Answer:
It learns the optimal weights and bias for classification.

---

20. What are limitations of perceptron learning law?

Answer:

- Works only for linearly separable data
- Cannot solve XOR problem
- Single-layer architecture

Procedure / Algorithm

Algorithm for Perceptron Learning Law

Step 1:

Initialize:

- weights
- bias
- learning rate

Step 2:

Provide training inputs and target outputs.

Step 3:

Calculate weighted sum:

$$Y_{in} = w_1 x_1 + w_2 x_2 + b$$

Step 4:

Apply activation function.

If:

$$Y_{in} \geq 0$$

output = 1

Else output = 0.

Step 5:

Compare predicted output with target output.

Step 6:

If error exists, update weights using:

$$w_{new} = w_{old} + \eta(t - y)x$$

Step 7:

Repeat for all training samples.

Step 8:

Continue training until classification error becomes zero or minimum.

Important Viva Cross Questions

Why is XOR not solved by perceptron?

Answer:
Because XOR data is not linearly separable and cannot be divided using a single straight line.

What is the difference between decision boundary and decision region?

Answer:
Decision boundary is the separating line, while decision regions are the areas formed after separation.

What is supervised learning in perceptron?

Answer:
The perceptron learns using labeled input-output training data.

Very Important Graph-Based Question

What does the graph show in this practical?

Answer:
The graph shows:

- training data points
- different classes
- decision boundary separating the classes

Quick Revision Line

Perceptron Learning Law updates weights iteratively to create a decision boundary that separates different classes correctly.

5

Statement 5: Bidirectional Associative Memory (BAM) with Two Pairs of Vectors

This viva mainly focuses on:

- BAM network
- Associative memory
- Bidirectional mapping
- Hebbian learning
- Recall process

---

1. What is BAM?

Answer:
Bidirectional Associative Memory (BAM) is a recurrent neural network used to store and recall associated pattern pairs.

---

2. Why is it called bidirectional?

Answer:
Because information can flow in both directions:

- input to output
- output to input

---

3. Who developed BAM?

Answer:
BAM was developed by Bart Kosko.

---

4. What is associative memory?

Answer:
Associative memory recalls stored output patterns when related input patterns are provided.

---

5. What are pattern pairs in BAM?

Answer:
Pattern pairs are input-output vector pairs stored in the BAM network.

Example:

$$(X_1, Y_1), (X_2, Y_2)$$

---

6. What type of network is BAM?

Answer:
BAM is a heteroassociative recurrent neural network.

---

7. What is heteroassociative memory?

Answer:
It associates one pattern with a different pattern.

Example:

$$X \to Y$$

8. What is autoassociative memory?

Answer:
Autoassociative memory recalls the same pattern that was given as input.

Example:

$$X \rightarrow X$$

---

9. What learning rule is used in BAM?

Answer:
Hebbian learning rule is used.

---

10. What is Hebbian learning?

Answer:
Hebbian learning states:
"Neurons that fire together wire together."

Weights are updated based on correlation between neurons.

---

11. How is weight matrix calculated in BAM?

Answer:
Weight matrix is obtained using outer product of input and output vectors.

$$W = X^T Y$$

For multiple pairs:

$$W = \sum X_i^T Y_i$$

---

12. What is the role of weight matrix?

Answer:
Weight matrix stores associations between input and output patterns.

---

13. What are bipolar vectors?

Answer:
Bipolar vectors use values:

$$-1 \text{ and } +1$$

instead of 0 and 1.

---

14. Why are bipolar values preferred in BAM?

Answer:
They simplify mathematical calculations and improve convergence.

---

15. What is recall in BAM?

Answer:
Recall means retrieving stored output patterns from given input patterns.

---

16. What is convergence in BAM?

Answer:
Convergence occurs when outputs stop changing and stable patterns are obtained.

---

17. Is BAM supervised or unsupervised?

Answer:
BAM mainly uses unsupervised Hebbian learning.

---

18. What are applications of BAM?

Answer:
Applications include:

- Pattern recognition
- Image association
- Memory retrieval
- Signal processing

---

19. Difference between BAM and Hopfield Network?

Answer:

- BAM stores associated pattern pairs
- Hopfield network stores single patterns

---

20. What are limitations of BAM?

Answer:

- Limited storage capacity
- May produce spurious states
- Noise can affect recall accuracy

Procedure / Algorithm

Algorithm for BAM

Step 1:

Define bipolar input-output vector pairs.

Example:

$$(X_1, Y_1), (X_2, Y_2)$$

Step 2:

Convert vectors into bipolar form:

$$0 \rightarrow -1$$

Step 3:

Compute outer product for each pair:

$$W = X^T Y$$

Step 4:

Add all outer products to form final weight matrix.

Step 5:

Provide input vector to BAM.

Step 6:

Compute output using:

$$Y = sgn(XW)$$

Step 7:

Feed output back to input layer:

$$X = sgn(YW^T)$$

Step 8:

Repeat until stable output is obtained.

Important Cross Questions

Why is BAM called recurrent network?

Answer:
Because outputs are fed back into the network repeatedly until convergence occurs.

What is a stable state in BAM?

Answer:
A stable state is reached when outputs no longer change after iterations.

Why is feedback important in BAM?

Answer:
Feedback helps refine recalled patterns and achieve stable memory retrieval.

Very Important Viva Question

Difference between feedforward and recurrent network?

Answer:

- Feedforward network: data moves only in one direction
- Recurrent network: data can flow back through feedback connections

---

Quick Revision Line

BAM is a bidirectional recurrent neural network that stores and recalls associated pattern pairs using Hebbian learning.

6

Statement 6: Create Neural Network Architecture from Scratch for Multi-Class Classification

This viva mainly focuses on:

- ANN architecture
- Multi-class classification
- Hidden layers
- ReLU
- Output layer
- Forward propagation
- Optimization

---

1. What is an Artificial Neural Network (ANN)?

Answer:
ANN is a computational model inspired by the human brain, consisting of interconnected neurons used for learning patterns from data.

---

2. What is meant by "from scratch"?

Answer:
It means implementing the neural network manually using programming logic instead of using prebuilt deep learning libraries.

---

3. What is multi-class classification?

Answer:
Multi-class classification means classifying data into more than two categories.

Example:

- digits 0–9
- animal classes
- flower species

---

4. What are the layers in ANN?

Answer:
Main layers are:

- Input layer
- Hidden layer(s)
- Output layer

---

5. What is the role of input layer?

Answer:
Input layer receives features from the dataset and passes them to the network.

---

6. What is a hidden layer?

Answer:
Hidden layers perform computations and extract important patterns from data.

---

7. Why are hidden layers important?

Answer:
They help the network learn complex and non-linear relationships.

---

8. Why are 100 neurons used in hidden layer?

Answer:
More neurons increase learning capability and help capture complex patterns.

---

9. What is ReLU activation function?

Answer:
ReLU outputs positive values directly and converts negative values to zero.

$$f(x) = \max(0, x)$$

---

10. Why is ReLU commonly used?

Answer:
ReLU is computationally efficient and reduces vanishing gradient problems.

---

11. What is the output layer?

Answer:
The output layer produces final predictions or class probabilities.

---

12. Why are multiple neurons used in output layer?

Answer:
Because multi-class classification requires one neuron for each class.

---

13. What activation is used in output layer for multi-class classification?

Answer:
Softmax activation function is commonly used.

---

14. What does Softmax do?

Answer:
Softmax converts outputs into probability values whose total sum equals 1.

---

15. What is forward propagation?

Answer:
Forward propagation is the process of passing inputs through layers to compute outputs.

---

16. What is backpropagation?

Answer:
Backpropagation updates weights by propagating error backward through the network.

---

17. What is an optimizer?

Answer:
Optimizer adjusts weights to minimize error during training.

---

18. Name common optimization algorithms.

Answer:
Common optimizers are:

- Gradient Descent
- SGD
- Adam
- RMSProp

---

19. Which optimizer is mostly used?

Answer:
Adam optimizer is widely used because it converges faster and works efficiently.

---

20. What is loss function?

Answer:
Loss function measures the difference between predicted and actual outputs.

---

21. Which loss function is used for multi-class classification?

Answer:
Categorical Cross-Entropy is commonly used.

---

22. What is an epoch?

Answer:
One complete pass through the entire training dataset is called an epoch.

---

23. What is overfitting?

Answer:
Overfitting occurs when the model memorizes training data and performs poorly on new data.

---

24. How can overfitting be reduced?

Answer:
Overfitting can be reduced using:

- Dropout
- Regularization
- More training data

---

25. What is threshold value?

Answer:
Threshold determines the minimum probability needed to assign a class label.

Procedure / Algorithm

Algorithm for ANN from Scratch

Step 1:

Load dataset for multi-class classification.

Step 2:

Initialize:

- weights
- biases
- learning rate

Step 3:

Create architecture:

- Input layer
- Hidden layer(s) with ReLU
- Output layer with Softmax

Step 4:

Perform forward propagation.

Hidden layer:

$$Z = WX + b$$

Apply ReLU activation.

Step 5:

Compute output probabilities using Softmax.

Step 6:

Calculate loss using cross-entropy.

Step 7:

Perform backpropagation to compute gradients.

Step 8:

Update weights using optimizer.

Step 9:

Repeat training for multiple epochs.

Step 10:

Evaluate model accuracy on test data.

Important Cross Questions

Why do we need hidden layers?

Answer:
Without hidden layers, the network behaves like a linear classifier and cannot solve complex problems.

Difference between binary and multi-class classification?

Answer:

- Binary classification → 2 classes
- Multi-class classification → more than 2 classes

Why is Softmax preferred for multi-class classification?

Answer:
Because it produces probability distribution across all classes.

Very Important Viva Question

Difference between ANN and Deep Learning?

Answer:
Deep learning is an advanced form of ANN with many hidden layers capable of learning highly complex patterns.

Quick Revision Line

ANN for multi-class classification uses hidden layers with ReLU and an output layer with Softmax to classify inputs into multiple categories.

I think 7th is almost same as 6th so give me only the part which is new of 7th . dont give whole

Statement 7: ANN Training using Forward Propagation & Backpropagation

(Only New Important Viva Points)

---

1. What is Forward Propagation?

Answer:
Forward propagation is the process where input data moves from input layer to output layer to generate predictions.

---

2. What happens during forward propagation?

Answer:
The network computes weighted sums and applies activation functions layer by layer to produce output.

---

3. Formula used in forward propagation?

$$Z = WX + b$$

Then activation function is applied.

---

4. What is Backpropagation?

Answer:
Backpropagation is the process of sending error backward through the network to update weights and reduce loss.

---

5. Why is backpropagation important?

Answer:
It helps the neural network learn by minimizing prediction error.

---

6. What is gradient descent?

Answer:
Gradient descent is an optimization algorithm used to minimize loss by updating weights in the negative gradient direction.

---

7. Weight update formula in backpropagation?

$$W_{new} = W_{old} - \eta \frac{\partial L}{\partial W}$$

---

8. What is loss propagation?

Answer:
The calculated error at output layer is propagated backward to hidden layers during backpropagation.

---

9. What is chain rule in backpropagation?

Answer:
Chain rule is used to compute gradients of loss with respect to weights layer by layer.

---

10. What is gradient?

Answer:
Gradient indicates how much the loss changes with respect to weights.

---

11. What is vanishing gradient problem?

Answer:
In deep networks, gradients become extremely small, slowing or stopping learning.

---

12. Which activation helps reduce vanishing gradient?

Answer:
ReLU helps reduce vanishing gradient problem.

Extra Algorithm Part (New)

Training Steps Specifically for Backpropagation

1. Perform forward propagation
2. Compute loss/error
3. Calculate gradients
4. Propagate error backward
5. Update weights and biases
6. Repeat for multiple epochs

Most Important Difference

Forward Propagation vs Backpropagation

Forward Propagation	Backpropagation
Computes output	Updates weights
Input → Output	Output → Hidden layers
Used for prediction	Used for learning

Quick Revision Line

Forward propagation generates predictions, while backpropagation updates weights to minimize error.

8

Statement 8: Backpropagation Network for XOR Function

This viva mainly focuses on:

- XOR problem
- Backpropagation
- Multi-layer neural network
- Non-linear separability

---

1. What is XOR function?

Answer:
XOR (Exclusive OR) gives output 1 when inputs are different and 0 when inputs are same.

---

2. XOR Truth Table

X1	X2	Output
0	0	0
0	1	1
1	0	1
1	1	0

---

3. Why is XOR important in neural networks?

Answer:
XOR demonstrates that single-layer perceptrons cannot solve non-linearly separable problems.

---

4. Why can't perceptron solve XOR?

Answer:
Because XOR data cannot be separated using a single straight-line decision boundary.

---

5. What type of problem is XOR?

Answer:
XOR is a non-linearly separable classification problem.

---

6. How is XOR solved in ANN?

Answer:
XOR is solved using a multi-layer neural network with hidden layers and backpropagation.

---

7. Why are hidden layers required for XOR?

Answer:
Hidden layers help create non-linear decision boundaries needed to solve XOR.

---

8. What learning algorithm is used?

Answer:
Backpropagation algorithm is used.

---

9. What activation functions are commonly used?

Answer:
Sigmoid or ReLU activation functions are commonly used.

---

10. What is the role of backpropagation in XOR?

Answer:
Backpropagation adjusts weights iteratively to minimize XOR classification error.

---

11. Architecture commonly used for XOR?

Answer:

- 2 input neurons
- 1 hidden layer
- 1 output neuron

---

12. Why is XOR considered a benchmark problem?

Answer:
Because it proves the necessity of hidden layers in neural networks.

---

13. What happens during training?

Answer:
The network performs forward propagation, computes error, and updates weights using backpropagation.

---

14. What loss function is commonly used?

Answer:
Mean Squared Error (MSE) or Binary Cross-Entropy.

---

15. What is convergence in XOR training?

Answer:
Convergence occurs when the network predicts all XOR outputs correctly with minimum error.

Procedure / Algorithm

XOR using Backpropagation Algorithm

Step 1:

Initialize:

- weights
- biases
- learning rate

Step 2:

Provide XOR inputs and target outputs.

Step 3:

Perform forward propagation.

Compute:

$$Z = WX + b$$

Apply activation function.

Step 4:

Calculate output error.

Step 5:

Perform backpropagation.

Compute gradients and propagate error backward.

Step 6:

Update weights:

$$W_{new} = W_{old} - \eta \frac{\partial L}{\partial W}$$

Step 7:

Repeat training for multiple epochs until error becomes minimum.

Important Cross Questions

What is non-linear separability?

Answer:
When classes cannot be separated using a single straight line, they are non-linearly separable.

Which network first solved XOR successfully?

Answer:
Multi-layer perceptron (MLP) with backpropagation solved XOR.

Why is XOR famous in ANN history?

Answer:
Because failure of perceptrons on XOR highlighted the need for multi-layer neural networks.

Very Important Viva Question

Difference between AND and XOR problems?

AND	XOR
Linearly separable	Non-linearly separable
Solved by perceptron	Requires hidden layer

Quick Revision Line

XOR is a non-linearly separable problem solved using multi-layer neural networks and backpropagation.

9

Statement 9: Backpropagation Feed-Forward Neural Network

(Only New Viva Points compared to Statement 7 & 8)

---

1. What is a Feed-Forward Neural Network?

Answer:
A Feed-Forward Neural Network (FFNN) is a neural network in which data flows only in one direction—from input layer to output layer.

---

2. Why is it called feed-forward?

Answer:
Because there are no feedback or recurrent connections; information moves only forward.

---

3. What are the layers in FFNN?

Answer:

- Input layer
- Hidden layer(s)
- Output layer

---

4. Difference between Feedforward and Recurrent Network?

Feedforward Network	Recurrent Network
One-way data flow	Feedback connections exist
No memory	Has memory
Simpler architecture	More complex

---

5. What is the role of hidden layers in FFNN?

Answer:
Hidden layers extract features and learn complex relationships from data.

---

6. Why is backpropagation used in FFNN?

Answer:
Backpropagation trains the network by minimizing prediction error through weight updates.

---

7. What type of learning is used in FFNN?

Answer:
Supervised learning is commonly used.

---

8. What are applications of FFNN?

Answer:

- Pattern recognition
- Classification
- Prediction
- Image recognition

---

9. What are advantages of FFNN?

Answer:

- Simple architecture
- Easy training

- Good for classification problems

---

10. Limitation of FFNN?

Answer:
It cannot remember previous inputs because it has no feedback mechanism.

Extra Procedure Part

Flow of FFNN

Input Layer
→ Hidden Layer(s)
→ Output Layer

(No backward connections)

Important Cross Question

Is CNN a feed-forward network?

Answer:
Yes. CNN is also a type of feed-forward neural network.

Quick Revision Line

Feed-forward neural network passes data only in the forward direction and is commonly trained using backpropagation.

10

Statement 10: Design a Hopfield Network which stores 4 vectors

This viva mainly focuses on:

- Hopfield Network
- Associative memory
- Energy function
- Stable states
- Pattern storage

---

1. What is Hopfield Network?

Answer:
Hopfield Network is a recurrent neural network used for storing and recalling patterns using associative memory.

---

2. Who developed Hopfield Network?

Answer:
Hopfield Network was developed by John Hopfield.

---

3. What type of network is Hopfield Network?

Answer:
It is a recurrent autoassociative neural network.

---

4. What is autoassociative memory?

Answer:
Autoassociative memory recalls the same pattern that was given during training.

Example:

$$X \to X$$

---

5. Why is Hopfield Network called recurrent?

Answer:
Because neurons are connected with feedback connections.

---

6. What are stable states?

Answer:
Stable states are stored patterns where neuron outputs stop changing.

---

7. What is the main purpose of Hopfield Network?

Answer:
Its main purpose is pattern storage and pattern recall.

---

8. What learning rule is used in Hopfield Network?

Answer:
Hebbian learning rule is commonly used.

---

9. How are weights calculated?

Answer:
Weights are calculated using outer products of stored vectors.

$$W = \sum X_i X_i^T$$

---

10. Why are diagonal weights zero?

Answer:
A neuron should not connect to itself, so diagonal weights are set to zero.

---

11. What values are generally used in Hopfield Network?

Answer:
Bipolar values:

$$-1, \ +1$$

---

12. What is energy function in Hopfield Network?

Answer:
Energy function measures the stability of the network. During recall, energy decreases until a stable state is reached.

---

13. Why is energy minimization important?

Answer:
It helps the network converge to stored stable patterns.

---

14. What is convergence in Hopfield Network?

Answer:
Convergence occurs when neuron states no longer change.

---

15. What are spurious states?

Answer:
Spurious states are unwanted stable states not originally stored in the network.

---

16. What are applications of Hopfield Network?

Answer:
Applications include:

- Pattern recognition
- Image restoration
- Error correction
- Memory retrieval

---

17. Difference between BAM and Hopfield Network?

BAM	Hopfield
Bidirectional association	Same pattern recall
Heteroassociative	Autoassociative
Two layers	Single layer

---

18. Can Hopfield Network handle noisy inputs?

Answer:
Yes. It can recover stored patterns from partially corrupted inputs.

Procedure / Algorithm

Algorithm for Hopfield Network

Step 1:

Define 4 bipolar vectors to store.

Step 2:

Convert vectors into bipolar form:

$$0 \rightarrow -1$$

Step 3:

Compute weight matrix using:

$$W = \sum X_i X_i^T$$

Step 4:

Set diagonal elements of weight matrix to zero.

Step 5:

Provide test input pattern.

Step 6:

Update neuron outputs using weighted sums.

Step 7:

Repeat updates until stable state is reached.

Step 8:

Final stable output gives recalled pattern.

Important Cross Questions

Why are bipolar vectors preferred?

Answer:
Bipolar vectors simplify mathematical operations and improve network stability.

What is the difference between stable state and spurious state?

Answer:

- Stable state → correct stored pattern
- Spurious state → incorrect unwanted pattern

Why is Hopfield called content-addressable memory?

Answer:
Because stored patterns can be recalled using partial or noisy input content.

Very Important Viva Question

What is the limitation of Hopfield Network?

Answer:

- Limited storage capacity
- May generate spurious states
- Performance decreases when many patterns are stored

Quick Revision Line

Hopfield Network is a recurrent autoassociative neural network used for storing and recalling patterns through energy minimization.

11

Statement 11: CNN Object Detection & Performance Evaluation Metrics

This viva mainly focuses on:

- CNN
- Object detection
- Bounding boxes
- Evaluation metrics

- Deep learning in computer vision

---

1. What is CNN?

Answer:
CNN (Convolutional Neural Network) is a deep learning model mainly used for image processing and computer vision tasks.

---

2. Why is CNN used for images?

Answer:
CNN automatically extracts important image features like edges, textures, and shapes.

---

3. What is object detection?

Answer:
Object detection identifies objects in an image and locates them using bounding boxes.

---

4. Difference between image classification and object detection?

Image Classification	Object Detection
Predicts object class only	Predicts class + location
One label	Multiple objects possible

---

5. What is a bounding box?

Answer:
A bounding box is a rectangular box used to locate an object in an image.

---

6. Main layers in CNN?

Answer:

- Convolution layer
- Pooling layer
- Fully connected layer

---

7. What is convolution layer?

Answer:
Convolution layer extracts features from images using filters/kernels.

---

8. What is a filter/kernel?

Answer:
A filter is a small matrix used to detect image features like edges and textures.

---

9. What is pooling?

Answer:
Pooling reduces image dimensions and computation while preserving important features.

---

10. Types of pooling?

Answer:

- Max pooling
- Average pooling

---

11. What is ReLU in CNN?

Answer:
ReLU introduces non-linearity and improves training speed.

$$f(x) = \max(0, x)$$

---

12. What is feature extraction?

Answer:
Feature extraction means automatically identifying important patterns from images.

---

13. What is flattening?

Answer:
Flattening converts 2D feature maps into 1D vectors before fully connected layers.

14. What are fully connected layers?

Answer:
Fully connected layers perform final classification based on extracted features.

---

15. What are common object detection algorithms?

Answer:

- R-CNN
- Fast R-CNN
- Faster R-CNN
- YOLO
- SSD

---

16. Why is YOLO popular?

Answer:
YOLO (You Only Look Once) performs real-time object detection with high speed.

---

17. What dataset is commonly used in object detection?

Answer:
COCO and Pascal VOC datasets are commonly used.

PERFORMANCE EVALUATION METRICS

18. What is accuracy?

Answer:
Accuracy is the ratio of correct predictions to total predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

---

19. What is precision?

Answer:
Precision measures how many predicted positive detections are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

---

20. What is recall?

Answer:
Recall measures how many actual objects are correctly detected.

$$Recall = \frac{TP}{TP + FN}$$

---

21. What is F1-score?

Answer:
F1-score is the harmonic mean of precision and recall.

---

22. What is IoU (Intersection over Union)?

Answer:
IoU measures overlap between predicted bounding box and actual bounding box.

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union}$$

---

23. Why is IoU important?

Answer:
IoU determines how accurately the object is localized.

---

24. What is mAP?

Answer:
mAP (mean Average Precision) is the standard metric used to evaluate object detection models.

---

25. What is overfitting in CNN?

Answer:
Overfitting occurs when the model performs well on training data but poorly on unseen data.

Procedure / Algorithm

CNN Object Detection Flow

Step 1:

Load image dataset.

Step 2:

Apply convolution layers for feature extraction.

Step 3:

Apply ReLU activation.

Step 4:

Apply pooling to reduce dimensions.

Step 5:

Flatten feature maps.

Step 6:

Use fully connected layers for prediction.

Step 7:

Generate:

- class labels
- bounding boxes

Step 8:

Evaluate model using:

- Accuracy
- Precision
- Recall
- IoU
- mAP

Important Cross Questions

Why is CNN better than traditional ANN for images?

Answer:
CNN automatically extracts spatial image features and requires fewer parameters.

What is the role of pooling?

Answer:
Pooling reduces computational complexity and helps prevent overfitting.

Why is object detection harder than image classification?

Answer:
Because it requires both object classification and object localization.

Very Important Viva Question

Difference between YOLO and R-CNN?

YOLO	**R-CNN**
Faster	Slower
Real-time detection	High accuracy
Single-stage detector	Multi-stage detector

Quick Revision Line

CNN performs object detection by extracting image features and locating objects using bounding boxes, evaluated using metrics like IoU and mAP.

12

Statement 12: Train Neural Network using TensorFlow/PyTorch & Logistic Regression Evaluation

This viva mainly focuses on:

- TensorFlow
- PyTorch
- Logistic Regression
- Training process
- Deep learning frameworks

---

1. What is TensorFlow?

Answer:
TensorFlow is an open-source deep learning framework developed by Google for building and training neural networks.

---

2. What is PyTorch?

Answer:
PyTorch is an open-source deep learning framework developed by Meta used for deep learning and research applications.

---

3. Difference between TensorFlow and PyTorch?

TensorFlow	PyTorch
More production-oriented	More research-oriented
Static + dynamic graphs	Dynamic computation graph
Widely used in deployment	Easier debugging

---

4. What is Logistic Regression?

Answer:
Logistic Regression is a supervised learning algorithm used for binary classification problems.

---

5. Why is it called regression if used for classification?

Answer:
Because it predicts probability values using a regression-based mathematical model.

---

6. What activation function is used in Logistic Regression?

Answer:
Sigmoid activation function is used.

$$f(x) = \frac{1}{1 + e^{-x}}$$

---

7. Why is sigmoid used?

Answer:
Because sigmoid converts outputs into probability values between 0 and 1.

---

8. What is binary classification?

Answer:
Binary classification means classifying data into two classes only.

Example:

- Spam / Not Spam
- Even / Odd

---

9. Steps to train neural network in TensorFlow/PyTorch?

Answer:

- Load dataset
- Create model
- Define loss function

- Select optimizer
- Train model
- Evaluate accuracy

---

10. What is a tensor?

Answer:
Tensor is a multidimensional data structure used in TensorFlow and PyTorch.

---

11. What is an optimizer?

Answer:
Optimizer updates weights to minimize loss during training.

---

12. Which optimizer is commonly used?

Answer:
Adam optimizer is commonly used because of faster convergence.

---

13. What is loss function in Logistic Regression?

Answer:
Binary Cross-Entropy loss is commonly used.

---

14. What is model training?

Answer:
Training is the process of adjusting weights using data to improve predictions.

---

15. What is evaluation in machine learning?

Answer:
Evaluation measures model performance on unseen data.

---

16. Common evaluation metrics for Logistic Regression?

Answer:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

---

17. What is a confusion matrix?

Answer:
Confusion matrix shows correct and incorrect predictions using TP, TN, FP, and FN.

---

18. What is accuracy?

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Answer:
Accuracy measures overall correct predictions.

---

19. What is precision?

$$Precision = \frac{TP}{TP + FP}$$

Answer:
Precision measures correctness of positive predictions.

---

20. What is recall?

$$Recall = \frac{TP}{TP + FN}$$

Answer:
Recall measures how many actual positives are correctly predicted.

Procedure / Algorithm

Neural Network Training in TensorFlow/PyTorch

Step 1:

Import required libraries.

Step 2:

Load and preprocess dataset.

Step 3:

Create neural network model.

Step 4:

Define:

- loss function
- optimizer

Step 5:

Train model using forward propagation and backpropagation.

Step 6:

Evaluate model using accuracy and other metrics.

Important Cross Questions

Why are TensorFlow and PyTorch popular?

Answer:
They simplify neural network implementation and provide GPU acceleration.

Which is easier for beginners?

Answer:
PyTorch is generally considered easier because of its dynamic computation graph.

Difference between Logistic Regression and ANN?

Logistic Regression	**ANN**
Simple linear model	Multi-layer model
Binary classification	Complex tasks
Less computational power	More powerful

Very Important Viva Question

Is Logistic Regression a deep learning algorithm?

Answer:
No. Logistic Regression is a machine learning algorithm, not a deep learning model.

Quick Revision Line

TensorFlow and PyTorch are deep learning frameworks used to train neural networks, while Logistic Regression is a binary classification algorithm using sigmoid activation.

13

Statement 13: TensorFlow/PyTorch Implementation of CNN

(Only New Important Viva Points)

---

1. What is implementation of CNN in TensorFlow/PyTorch?

Answer:
It means creating, training, and testing a CNN model using deep learning frameworks like TensorFlow or PyTorch.

---

2. Why are TensorFlow/PyTorch used for CNN?

Answer:
They provide built-in functions for layers, optimizers, GPU support, and efficient training.

---

3. What are the main layers used in CNN implementation?

Answer:

- Convolution layer

- ReLU activation
- Pooling layer
- Fully connected layer

---

4. What is Conv2D?

Answer:
Conv2D is a convolution layer used to extract spatial features from 2D images.

---

5. What is MaxPooling2D?

Answer:
MaxPooling2D reduces image dimensions by selecting maximum values from feature regions.

---

6. What is the role of Flatten layer?

Answer:
Flatten converts 2D feature maps into 1D vectors before classification.

---

7. What is dropout in CNN?

Answer:
Dropout randomly disables neurons during training to reduce overfitting.

---

8. What optimizer is commonly used in CNN?

Answer:
Adam optimizer is commonly used.

---

9. What loss function is used in CNN classification?

Answer:

- Binary Cross-Entropy → binary classification
- Categorical Cross-Entropy → multi-class classification

---

10. What datasets are commonly used in CNN practicals?

Answer:

- MNIST
- CIFAR-10
- Fashion-MNIST

---

11. Why is GPU important in CNN training?

Answer:
GPU performs parallel computations and significantly speeds up deep learning training.

---

12. What is batch size?

Answer:
Batch size is the number of training samples processed at one time before updating weights.

---

13. What is filter size in CNN?

Answer:
Filter size determines the dimensions of the kernel used for feature extraction.

Example:

$$3 \times 3,\ 5 \times 5$$

---

14. What is stride?

Answer:
Stride defines how many pixels the filter moves during convolution.

---

15. What is padding?

Answer:
Padding adds extra pixels around images to preserve dimensions after convolution.

Extra Procedure Part

CNN Implementation Flow

Step 1:

Load image dataset.

Step 2:

Create CNN architecture:

- Conv2D
- ReLU
- Pooling
- Flatten
- Dense layers

Step 3:

Compile model with:

- optimizer
- loss function
- metrics

Step 4:

Train CNN model on dataset.

Step 5:

Evaluate accuracy on test data.

Important Cross Questions

Why is ReLU used in CNN?

Answer:
ReLU improves training speed and reduces vanishing gradient problems.

Why is pooling important?

Answer:
Pooling reduces dimensions and computational complexity.

Difference between ANN and CNN?

ANN	CNN
Fully connected	Uses convolution
Less suitable for images	Best for images
More parameters	Parameter sharing

Very Important Viva Question

Why is CNN better than ANN for image tasks?

Answer:
CNN preserves spatial relationships and automatically extracts image features.

Quick Revision Line

CNN implementation in TensorFlow/PyTorch involves convolution, pooling, flattening, and dense layers for image classification tasks.

14

Statement 14: MNIST Handwritten Character Detection using PyTorch, Keras and TensorFlow

This viva mainly focuses on:

- MNIST dataset
- Handwritten digit recognition
- Deep learning frameworks
- CNN basics
- Image classification

---

1. What is MNIST?

Answer:
MNIST is a dataset of handwritten digit images from 0 to 9 used for training and testing machine learning models.

---

2. What does MNIST stand for?

Answer:
MNIST stands for Modified National Institute of Standards and Technology dataset.

---

3. How many classes are present in MNIST?

Answer:
There are 10 classes representing digits 0 to 9.

---

4. What is the image size in MNIST?

Answer:
Each image is:

$$28 \times 28$$

grayscale pixels.

---

5. Why is MNIST popular?

Answer:
MNIST is simple, standardized, and widely used for learning deep learning concepts.

---

6. What type of problem is MNIST?

Answer:
MNIST is a multi-class image classification problem.

---

7. What frameworks are used in this practical?

Answer:

- TensorFlow
- Keras
- PyTorch

---

8. What is Keras?

Answer:
Keras is a high-level deep learning API that runs on top of TensorFlow.

---

9. Why are CNNs commonly used for MNIST?

Answer:
CNNs automatically extract image features and provide high accuracy in digit recognition.

---

10. What preprocessing is done on MNIST images?

Answer:
Images are normalized by dividing pixel values by 255 to scale values between 0 and 1.

---

11. What activation function is commonly used?

Answer:
ReLU is used in hidden layers.

$$f(x) = \max(0, x)$$

---

12. What activation is used in output layer?

Answer:
Softmax activation is used for multi-class classification.

---

13. What is Softmax?

Answer:
Softmax converts outputs into probability values whose sum equals 1.

---

14. What loss function is commonly used?

Answer:
Categorical Cross-Entropy loss is commonly used.

---

15. What optimizer is commonly used?

Answer:
Adam optimizer is commonly used for faster convergence.

---

16. What is epoch in MNIST training?

Answer:
One complete pass through the entire MNIST dataset is called an epoch.

---

17. What is accuracy in MNIST?

Answer:
Accuracy measures how many handwritten digits are classified correctly.

---

18. Why is normalization important?

Answer:
Normalization improves training stability and convergence speed.

---

19. What is overfitting in MNIST?

Answer:
Overfitting occurs when the model memorizes training data and performs poorly on unseen digits.

---

20. How can overfitting be reduced?

Answer:
Using:

- Dropout
- More data
- Regularization

Procedure / Algorithm

MNIST Digit Recognition Flow

Step 1:

Load MNIST dataset.

Step 2:

Normalize image pixel values.

Step 3:

Build neural network/CNN model.

Step 4:

Add:

- convolution layers
- ReLU
- pooling
- dense layers

Step 5:

Compile model using:

- optimizer
- loss function
- accuracy metric

Step 6:

Train model on MNIST training data.

Step 7:

Test model on unseen handwritten digits.

Step 8:

Predict digit classes from output probabilities.

Important Cross Questions

Why is CNN preferred for handwritten digit recognition?

Answer:
CNN automatically extracts spatial features like edges and shapes from images.

Difference between TensorFlow and Keras?

Answer:
TensorFlow is a deep learning framework, while Keras is a high-level API built on TensorFlow.

Why is Softmax used in MNIST?

Answer:
Because MNIST has multiple classes (0–9), and Softmax provides probability distribution across classes.

Very Important Viva Question

Why is MNIST considered an introductory deep learning dataset?

Answer:
Because it is small, simple, easy to train, and suitable for learning neural network concepts.